Innovation in Sandbox - Group Project

Rebuilding Trust in Vehicle Servicing

Sector: Automotive, After-Sales Service & Vehicle Maintenance

From a problem inside the service bay to a tested, scale-ready idea

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WHERE WE LEFT OFF

The problem hasn't changed. Our understanding of it has deepened.

Sharp problem statement (Phase 1)

Urban Indian vehicle owners lose confidence in authorised service centres because advisors rarely explain diagnostic findings, cost changes, or delays as they happen. That gap pushes cost-sensitive, time-pressed customers toward unorganised local garages, despite the quality and safety risk.

Cycle time

Trust is built over months of visits, not one interaction.

Capital intensity

A physical-infrastructure business, not pure software.

Regulatory constraint

OEM warranty terms and data protection rules.

Failure cost

A wrong diagnosis is a safety issue, not just a bad review.

PHASE 2

21 ideas before we judged a single one

We ran 2 frameworks from class: the 6 Ideation Triggers, and Assumption Busting on our own category.

Frustrations

- A live WhatsApp ping every time the job moves to a new stage: received, diagnosed, approved, in repair, ready
- A one-tap approve or decline button attached to every extra cost line, sent before the work starts rather than billed after
- A digital job card the customer can open on their own phone, showing what the technician is doing right now instead of what was done 3 hours ago
- A plain SMS or automated voice-call fallback for customers who do not use WhatsApp or do not own a smartphone

Anomalies

- Why does the final bill only appear at pickup? A running bill that updates as each part or labour line is added
- Why does the advisor call only when there is extra cost, never to say things are still on track? Short "nothing to report, still on schedule" pings between the real updates
- Why can a customer track a food delivery rider in real time but not their own car's repair? A short diagnostic photo or video clip attached to the job card at the point it is taken

Confluence

- Smartphone penetration plus UPI adoption: pay-as-you-approve micro-payments per repair stage instead of one lump sum at collection
- QR codes plus WhatsApp: scan a QR code at drop-off to open a live tracking thread tied to that specific job
- Video calling plus remote-work habits: an optional 2-minute video walkthrough of the vehicle with the technician before approving a major repair

Orthodoxies

- Ques: "the service advisor is the only channel."
  Ans: a simple chatbot handles routine status questions so the advisor is only pulled in for judgement calls
- Ques: "an estimate is given once, at the start."
  Ans: a tiered estimate like definite, likely, possible shown upfront, so a later revision feels expected rather than sprung on the customer

Extremes

- A customer who has to leave town right after drop-off: a fully remote approval flow with a digital signature
- A customer who has been overcharged before and no longer trusts any service centre: an optional independent photo-verification step for major repairs
- A first-time car owner who does not understand technical terms: plain-language notes attached to each diagnostic line, written for someone who has never opened a bonnet

Analogies

- From food delivery: a named-stage tracker, Received, Diagnosed, Approved, In Repair, Quality Check, Ready, the same idea as an order tracker
- From hospital patient portals: one login that stores every past visit, part replaced, and amount paid, so the customer is not starting from zero each time
- From courier tracking: a dynamic ready-by estimate that updates itself the moment a delay is known, instead of one static promise made at drop-off

PHASE 2

Scored against 4 lenses, not a popularity vote

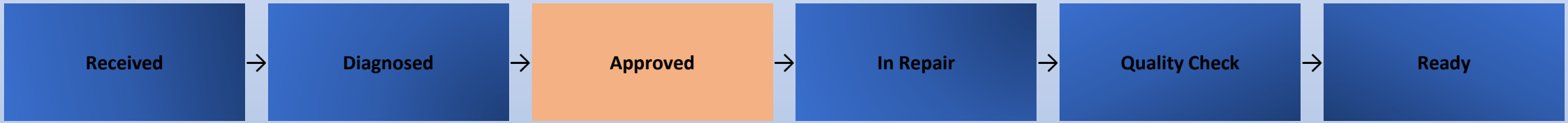
Shortlisted idea	Desire	Feasible	Viable	Novel	Total
Stage-wise WhatsApp status & approval flow	5	4	4	3	**16**
Dynamic running bill during the visit	4	3	3	3	13
Optional video walkthrough before major repairs	4	2	2	4	12
One login storing full service history	3	3	4	2	12
Independent third-party photo verification	3	2	2	4	11

The stage-wise WhatsApp flow won on desirability and feasibility, and folds in the strongest pieces of the other 4 ideas: the approve/decline action, the named-stage tracker, and the automatic status pings.

THE CONCEPT

The Concept

A WhatsApp thread, opened by QR code at drop-off, that pings the customer at every job-stage change and asks for an explicit tap to approve any cost beyond the original estimate.



Every extra cost line pauses the job until the customer taps Approve.

No new hardware | Runs on existing WhatsApp Business tooling | Human technician still writes every diagnosis

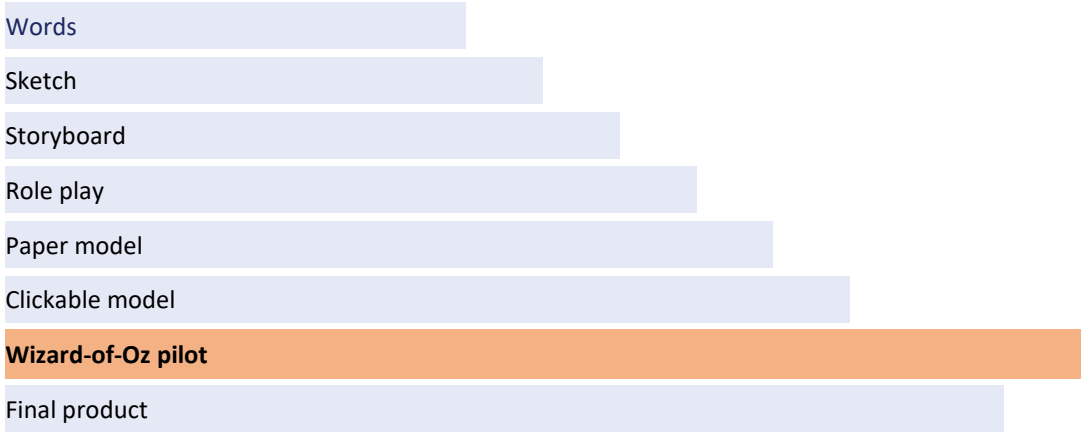
PHASE 3

The riskiest assumption isn't the idea. It's the behaviour.

Riskiest assumption

Vehicle owners will read, trust, and respond to a WhatsApp approval request from an unfamiliar number quickly enough that it feels faster than a phone call, not slower.

Prototype Ladder - where we chose to sit



A clickable mockup shows understanding, not behaviour under real pressure. A live bot integration is a bigger bet than an unproven assumption deserves. Wizard-of-Oz sits in between: it feels real to the customer, costs almost nothing to run.

PHASE 3

The Experiment Card, filled in before we ran a single test

ASSUMPTION	Customers respond to WhatsApp approval requests without a follow-up call, and trust the process at least as much as a phone call
TEST	Wizard-of-Oz pilot at a multi-brand garage and one authorised centre, across real service visits
METRIC	Response time | % approved without a follow-up call | 1-5 trust rating at pickup
SUCCESS THRESHOLD	≥70% respond within 30 minutes, no follow-up call | average trust rating ≥4
DECISION RULE	Persevere / Iterate / Pivot / Kill, set against the threshold above, not after seeing the result

PHASE 4

8 real vehicle owners. Not teammates. Not a survey.

6 at a multi-brand garage, 2 at an authorised centre, all mid-service on an ordinary booking. We watched what people did with the messages, then asked structured questions at pickup.

I Like

- Knowing the cost before it changed, not at the counter
- “Still on track” pings landed well

I Wish

- Messages from a saved name, not an unknown number, 1 almost read as spam
- A Hindi option

What If

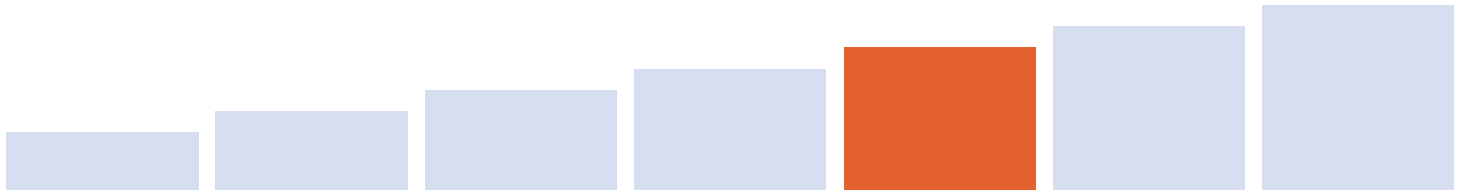
- The same thread could hold the next service reminder
- show the technician's name and photo

I Noticed

- Fastest replies came to messages with a rupee amount attached
- informational pings opened last

PHASE 4

Evidence, not opinion, and a decision to persevere



Our pilot landed around Trial: real usage in one visit, not yet repeat behaviour.

6 / 8

responded within 30 minutes, no follow-up call

4.1 / 5

average trust rating at pickup

1 / 8

missed the message entirely, unsaved number

PERSEVERE

Core assumption held. One fix carried forward: send from a verified, named business account, not a personal number.

PHASE 5

Scale Readiness Check

Desirability	Feasibility	Viability	Repeatability
Are customers repeatedly choosing it?	*Can we deliver it reliably?*	*Do the economics work?*	*Does it hold across contexts?*
Strong first-visit signal; no repeat-visit data yet	Yes, manually, at small scale. Needs a verified account and a light internal tool	Low cost per centre, but advisor time cost is not yet quantified	Only tested at two sites in one city; advisor discipline already varied between them

Desirability and feasibility are ready to move past pilot. Viability and repeatability need more evidence first.

PHASE 5

What stays constant. What has to change.

STAYS CONSTANT

- Tell the customer before asking them to pay, not after
- Every cost change needs an explicit tap to approve

HAS TO CHANGE

- Channel: WhatsApp by default, SMS/IVR fallback where smartphone use is lower
- Language and tone of templates, since our pilot only ran in English
- Warranty-disclosure wording, to match each OEM's dealership rules

WHERE THIS LEAVES US

1 tested idea, 3 open questions

This concept closes the exact communication gap identified in Phase 1, and a Wizard-of-Oz pilot with 8 real customers gave us evidence, not just opinion, that the core assumption holds.

1. More repeat-visit data before calling desirability proven
2. A clear read on advisor time cost before calling viability proven
3. A real data-retention policy before this touches customer numbers at scale

Design thinking is a loop, not a straight line and this is where the loop keeps going.